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Older adult frontal plane balance during 90 degree turns while walking

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This study examined how older adults regulated their balance during turns relevant to everyday mobility, towards informing future fall-risk mitigation approaches. Sixteen healthy older adults participated in this study. Balance biomechanics were quantified using whole-body optical motion capture. Three tasks were performed 10–14 times: straight line gait, 90˚ pre-planned left turns, and 90˚ late-cued left turns that were cued visually. Frontal plane angular momentum ranges and the minimum horizontal distance between the center of mass and lateral edge of the base of support (lateral distance) were compared across tasks using linear mixed models. Lateral distance minima were larger during straight-line gait than either turning task, and larger during late-cued than pre-planned turns. Group-level analysis showed that frontal plane angular momentum ranges were smaller during straight-line gait than during either turn task. Participant-specific analyses revealed that not all participants followed the group-level statistical findings for angular momentum, with some demonstrating the opposite behavior. To explore these diverse balance behaviors, preliminary explorations found significant associations between balance state extrema and baseline assessment scores (e.g., greater concern about falling was associated with smaller angular momentum ranges). Additionally, this study prompts more thoughtful interpretations of balance biomechanics when attempting to characterize an individual's fall-risk.

Keywords Angular momentum, Lateral distance, Older adult, Balance, Turn, Gait

Abbreviations

COM	Center of mass
BOS	Base of support
H_f	Frontal plane angular momentum
LD	Lateral distance
DGI	Dynamic Gait Index
MOCA	Montreal Cognitive Assessment
FES-I	International Fear Efficacy Scale
TUG	Timed Up and GO
Dual-TUG	Dual Timed Up and Go
PP	Pre-planned turns
LC	Late-cued turns
SLG	Straight-line gait

Falls are a leading cause of fatal and non-fatal injuries in older adults in the U.S[1]. Around one in three older adults fall each year[2], leading to a heavy quality of life impact and over $50 billion in medical costs annually[3]. Up to half of all walking steps taken daily are part of turns, depending on the environment[4]. Unfortunately, falls during turns were found to be 7.9 times more likely to result in a hip fracture than falls during straight line walking[5], and hip fractures in older adults lead to a severe decline in independence and quality of life[6,7]. Turning is a dynamic task that requires changing the body's facing and travel directions while keeping the center of mass (COM) controlled with respect to the base of support (BOS)[8–10].

Aging creates unique challenges that can inhibit effective turning while walking, and the strategies that are used to maintain balance during turning tend to change with age. Older adults have been shown to turn more

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slowly and use different movement strategies when compared with younger adults[11–13]. For example, Thigpen et al. (2000) showed that during 180° turns, older adults tended to take more steps to execute the turn in place of a pivot turn, which otherwise would rotate the body in one movement over the planted foot[12]. Similarly, Dixon et al. (2018) showed that, unlike young adults, older adults do not tend to "lean" into the new walking direction when performing a turn[13]. Although turning can be challenging for older adults and poses higher risk of falls and injury, there is minimal research focusing on how older adults control their balance during turning while walking.

This study focuses on frontal plane balance metrics. Previous studies indicate that the ability to control balance in the frontal plane is particularly important towards fall mitigation[14,15]. Gait studies have indicated that active control is necessary to adjust mediolateral balance in the frontal plane, but is not as demanding in the sagittal plane[16–19]. Additionally, mediolateral instability tends to be associated with an increased risk of falls and fall-related injuries for older adults[20–22]. When balance is lost in the lateral direction during single support, a "crossover" corrective step - stepping across the body - could be needed to prevent a fall. This is a more difficult corrective maneuver for older adults than losing balance medially during single support, or in the antero-posterior direction[20].

Angular momentum has been emphasized as a useful balance metric associated with fall-risk in older adults[23–27]. Frontal-plane angular momentum (H_f) is a measure of rotational speed of the body around the whole body's COM in the frontal-plane. Older adults tend to have a larger H_f range, representing H_f extrema, during straight-line gait than younger adults[23,28], which has been associated with a lesser ability to control H_f and a greater risk of falls[29,30]. Previous research with young adults has shown that H_f range tends to increase during turns when compared with straight-line gait[8]. However, this has not yet been explored in older adults. Additionally, as a measure in the speed-domain, H_f often reaches maximum values during gait when the body is relatively upright and nears zero at times that body sway changes its direction (i.e., at extrema of body tilt orientations). Thus, angular momentum values do not indicate the states of body alignment or orientation, but they may associate with extreme values of body sway if H_f is not adequately controlled (i.e., attenuated after an extreme value). H_f is predominantly controlled during gait by the moment or torque generated about the body's COM from the ground reaction forces footfall to footfall. For example, taking a wider step (given the same ground reaction force patterns) may increase the moment arm for that ground reaction force to act to arrest the body's rotation, thus avoiding an unstable body orientation and quickly attenuating any previous extreme H_f values[23,31]. Thus, this study aims to better understand the regulation of not only H_f used by older adults during turns, but also the relevant (and independent) control of the COM's alignment with its BOS.

An additional balance metric, lateral distance (LD), has been helpful in describing balance states during turns[8,32]. LD is the frontal-plane distance between the body's COM location to the closest lateral edge of the BOS. Therefore, it describes the proximity of the COM to the lateral BOS edge and is negative if the COM is lateral of the lateral BOS edge. Previous research with young adults and children found that LD minima tend to be smaller or more negative during turns as compared to straight-line gait[8,32]. This indicates that people can experience more complex balance states during turns, which often includes times when the COM is lateral to the BOS. LD during turning has not yet been explored with older adults, so this study aims to better understand how older adults regulate their balance states to complete a successful and safe turn.

Turning takes place in a variety of contexts in daily life. This study looks at 90˚ turns in two contexts: pre-planned 90˚ turns and late-cued 90˚ turns. Pre-planned turns mimic scenarios where one can prepare adequately with the steps leading up to the turn. Late-cued turns are meant to represent an unexpected sudden turn, where there is no preparation leading up to the turn. Prior research shows that late-cued turns can pose increased challenges to maintain balance while redirecting momentum in the new direction of travel[8,13,33,34]. Understanding the way in which these challenges are managed are relevant to understand for future fall mitigation practices.

Describing how H_f and LD are regulated in these two turn types provides important rotational and positional balance regulation contexts that may relate to clinical measures used to assess fall risk. Current clinical practice includes a variety of established balance, cognitive and psychosocial assessments to indicate fall risk in older adults[35–41]. Therefore, towards the goal of better understanding and improving fall mitigation in older adults, this study also preliminarily explores how clinical fall-risk assessments may associate with biomechanical frontal plane balance measures during turns.

The purpose of this study is to better understand how older adults regulate their frontal plane balance during 90 degree pre-planned and late-cued turns. Based on previous research with young adults, we hypothesized that older adults would have (1) larger range of H_f during both pre-planned and late-cued turns as compared to straight-line gait, and (2) larger range of H_f during late-cued turns as compared to pre-planned turns. We also hypothesized that older adults would have (3) smaller LD minima during pre-planned and late-cued turns as compared to straight-line gait, and (4) smaller LD minima during pre-planned turns as compared to late-cued turns.

Methods

Participants

This study included 16 healthy older adults (14 female, age 71.3 ± 4.93 years, mass 70.8 ± 11.5 kg, height 1.63 ± 0.078 m) who provided informed consent to voluntarily participate in this study. The protocol was approved by the Stevens Institute of Technology Institutional Review Board and carried out in accordance with relevant guidelines, regulations, and the Declaration of Helsinki. Informed consent was obtained from all participants. Participants self-reported that they: were free of lower extremity injuries or pain, were able to walk for a quarter of a mile unassisted, were able to hear without the use of a hearing assistive device, had no diagnosed balance deficit, and were not currently in a physical therapy program. The Dynamic Gait Index (DGI), an eight-item clinical tool used to assess balance and fall risk, was administered to determine potential balance deficits. All participants

scored 19 or greater on the DGI[42] as part of the inclusion criteria. The Montreal Cognitive Assessment (MoCA) was administered to rule out cognitive impairment, and all participants scored 23 or greater[36] as an additional inclusion criterion. Additionally, all participants demonstrated decisional ability via the University of California, San Diego Brief Assessment of Capacity to Consent, which was customized for this study[43].

All 16 participants performed straight-line gait and pre-planned turns, but only eight participants performed late-cued turns due to time limitations, as the protocol required at least one hour remaining to complete the late-cued turns. In addition, out of an abundance of caution about fall-risk, the study physical therapist recommended against one participant completing the late cued turns due to fatigue and concern about understanding the task, even though there was adequate time (participant nine in subsequent figures).

Experimental setup

Any time an older adult participant was in the lab, a physical therapist was present to guard their balance and provide verbal instructions following a printed script. All participants were screened for severe fall-risk and cognitive health as eligibility requirements. Although not an eligibility requirement, each participant self-reported their fall history over the prior 12 months[44] and completed the International Falls-Efficacy Scale (FES-I) to indicate their concern of falling during every-day activities[38]. Participants also completed the Timed Up and Go (TUG) and Dual Timed Up and Go (Dual-TUG) assessments[39,40].

A grocery store aisle intersection was simulated using a T-shaped walkway indicated with bright blue tape boundaries on the black floor (Fig. 1). The walkway was 0.91 m wide[45], including a 10 m straight-way with a 90° turn in the center leading to a 5 m straight-way, as previously described[8]. Vertical poles were located at the entry to the turning aisle (Fig. 1). A screen (2.03 m diagonal) at the end of the 5 m intersection mimicked a grocery aisle signage for the turns. Three tasks were performed in an order of increased challenge as a safety precaution: straight-line gait, followed by pre-planned turns, and then late-cued turns. There was a five-minute instructional period before each task followed by two practice trials. For each trial, a starting foot was randomly prescribed with a 50% chance for the left and right starting foot. Participants were asked to walk as if they were not in a rush, but people were behind them such that they were not able to stop abruptly.

Participants performed 10–14 trials of straight-line gait followed by 10–14 trials of pre-planned turns (Fig. 1). For straight-line gait, they were asked to imagine they were in a grocery store walking straight down an aisle at their regular pace. For pre-planned turns, they were told to pretend that they were in a familiar grocery store, and they were shopping for broccoli. They knew which aisle to turn into, and a large picture of broccoli was shown on the screen to indicate the grocery store signage. They approached the intersection and turned to walk towards the screen at the end of the walkway.

Finally, participants who had at least one hour remaining performed 20–24 trials of late-cued turns: 10–12 late-cued turn trials and 10–12 late-cued catch trials (Fig. 1C) that were randomly ordered. For this task, participants were told that they were in an unfamiliar grocery store looking for broccoli and, when they reached the intersection, they should look at the screen to determine if the broccoli was down that aisle or not, knowing that there was a 50% chance that it was in that aisle. Once the participants reached the intersection, a broccoli or a "no" symbol (red circle with a line through it) was shown on the screen. The cue was initiated manually

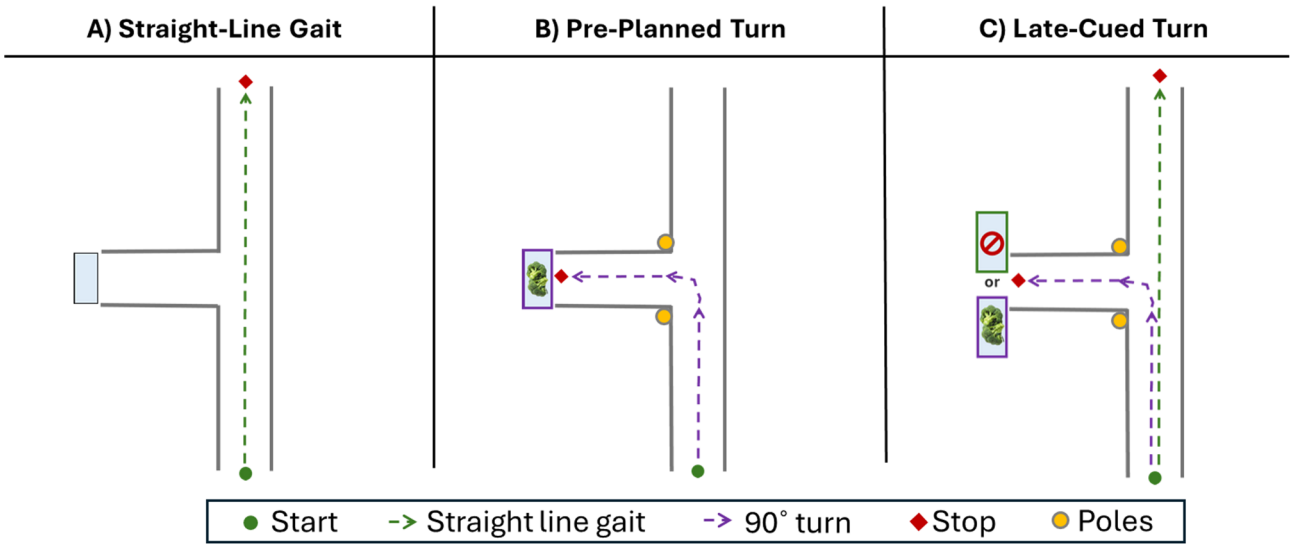

Fig. 1. Three walking tasks. **A** Straight-line gait for 10 m. **B** Pre-planned left turns, participants are told to pretend they are in a grocery store looking for broccoli. They walk 5 m to the intersection and then turn down the simulated grocery store aisle and walk another 5 m. **C** Late-cued turns, participants are told they are in an unfamiliar grocery store and only know if they should turn after they reach the intersection and look down the aisle at the cue on the screen. When green broccoli was displayed on the screen, it indicated that they should turn, and when a red circle with a line through it appeared, it indicated that they should continue walking straight. Poles were placed at the intersection corners to simulate an aisle.

by the researcher viewing when the foot position passed into the intersection on the motion capture computer screen, which displayed a video view aligned with the intersection for this purpose. If the broccoli was displayed, participants performed a 90° turn down the intersection (a late-cued turn trial). If the "no" symbol was shown, then the participant continued walking straight until the end of the 10 m walkway (a catch trial). The screen was blank until the participant reached the intersection (to avoid looking at the screen prior to the intersection).

Kinematic analysis

A 13-segment whole-body kinematic model[46] was built using optical motion capture data (250 fps, OptiTrack USA). Rigid tacking marker clusters were placed on the head, torso, pelvis, left and right upper arms, forearms, thighs, shanks, and feet. Virtual anatomical markers were identified (3D digitization) using a calibration pointing device (Probe Kit, Optitrack, Corvallis, OR), which were tracked through time and space using rigid tracking clusters (Fig. 2)[46]. All marker data were gap-filled and smoothed using a cubic spline filter (MATLAB 'csaps' function with the smoothing value set to 0.0005).

Frontal plane angular momentum

Whole-body COM location was computed using a weighted average of each segment's center of mass. Whole-body angular momentum (H) about the COM was computed as follows[8,23]:

$$\vec{H} = \sum_{i=1}^{n} [(\vec{r}_{COM_i} - \vec{r}_{COM_{wb}}) \times m_i (\vec{v}_{COM_i} - \vec{v}_{COM_{wb}}) + I_i \vec{\omega}_i]$$

where $\vec{r}_{COM_i}$ and $\vec{v}_{COM_i}$ represent the ith segment's COM position and velocity vectors respectively, $\vec{r}_{COM_{wb}}$ and $\vec{v}_{COM_{wb}}$ represent the whole-body COM position and velocity vectors respectively. m_i, I_i, and $\vec{\omega}_i$ represent the ith segment's mass, moment of inertia, and angular momentum vector respectively, and n represents the number of segments. The frontal plane component of the whole-body angular momentum was computed as the component in the frontal plane defined by the pelvis orientation. The vector normal to the frontal plane was the horizontal-plane component of the anterior axis of the pelvis (i.e., so any anterior or posterior pelvic tilt would not affect its computation). Thus, using the "right-hand rule", when viewed posteriorly, H_f is positive during clockwise body rotation (i.e., the body above the COM rotating rightward) and negative during counterclockwise body rotation (i.e., the body above the COM rotating leftward). The maxima values and range within the phase of interest (described shortly) were computed for each trial. To ease comparisons with previous research and across participants, H_f was normalized to a unitless form by dividing H_f by the product of each participant's height (l), mass, and $\sqrt{g * l}$, where g is gravitational acceleration[8,23].

Lateral distance

LD was calculated by determining the horizontal distance between the total body COM and the lateral edge of the BOS. The lateral direction was defined using the horizontal plane component of the pelvis's mediolateral axis. The distance along this vector from the COM to the leading foot's lateral edge was used as LD. The BOS was defined by the foot markers (i.e., all markers below the ankle in Fig. 2) below a height threshold established during the static calibration trial (when both feet were in contact with the ground) plus a 1 cm tolerance. As the COM approaches the lateral edge of the BOS, the LD value decreases. If the COM is lateral of the lateral

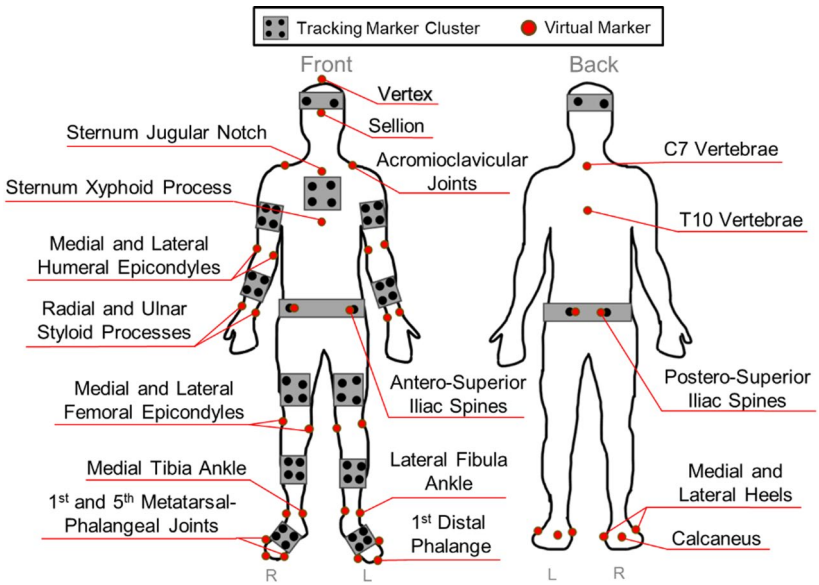

Fig. 2. Front and back of participant with tracking markers on the head, torso, pelvis, left and right upper arms, forearms, thighs, shanks, and feet. 3D digitized virtual markers marked by red dots and listed.

edge of the BOS, the LD value is negative. To ease comparisons with previous research and visualization across participants, LD was normalized by dividing LD by each participant's leg length[8]. Minima LD values were computed within the phase of interest for each trial.

Phases of interest

The phase of interest for straight-line gait was defined by the heel strike before and after the center 4 m of the walkway, to capture steady state walking. For the turning trials, the phase of interest was defined by a person-specific threshold determined by the average pelvis heading angle during straight-line gait plus or minus three times the standard deviation[8,47–49]. The turn phase began at the heel strike before the pelvis heading angle exceeded the threshold and ended at the heel strike after the pelvis heading angle returned below the threshold relative to the body's new direction of travel[8].

Additional contextual kinematic measures

Mean gait speed was computed during each task's phase of interest. Additional contextual kinematic computations included mean step and stride length, and stride width using the methods described by Huxham et al. (2006)[50], as well as mean stride duration, phase duration, and number of footfalls per each task's phase of interest.

Baseline assessments

In addition to the biomechanical balance metrics described, this study explores preliminary associations between clinical, cognitive and psychosocial assessments and balance during walking and turning. The DGI is an 8-item clinical tool used to assess balance and fall-risk, with higher scores indicating better balance[35]. The MOCA is a 16-item cognitive assessment, with higher scores indicating better cognitive abilities[36,37]. The FES-I is a self-administered 16-item questionnaire to determine fear of falling during daily activities, with higher scores indicating a greater fear of falling[38]. The TUG is a clinical assessment that measures the amount of time it takes to stand, walk 10 feet, return to the chair, and sit back down[39]. The Dual-TUG includes an additional task of counting backwards by three while performing the TUG assessment. For both, faster times indicate better balance and gait[40,41].

Statistical analyses

Differences in H_f, LD, and kinematic measures across straight-line gait and turning tasks were examined using linear mixed models that included random intercepts for study participants, random slopes for trial number nested within study task and study participants, and fixed effects for study task. Mixed models were chosen because they allowed for the appropriate handling of repeated measurements within study participants and innately maximize the information from subjects with missing observations. Pairwise comparisons between study tasks were estimated within the regression models via orthogonal contrasts. Residuals were examined for all models to ensure that all assumptions were satisfied. The Holm test was used to correct for multiple comparisons and to maintain a two-tailed familywise alpha at 0.05 across hypotheses tested. An adjusted p-value < 0.05 was used to determine statistical significance[51].

Exploratory analyses were conducted (1) to examine whether assessment scores were related to H_f or LD and (2) to examine whether assessment scores moderate the relationship between study task and H_f or LD. Mixed models that included random intercepts for study participants and random slopes for trial number nested within study task and study participants were used for both exploratory aims. To explore whether assessment scores related to H_f or LD, the mixed model also included a fixed effect for assessment score. To explore if the assessments scores moderate relationships between study task and H_f or LD, the mixed model also included fixed effects for study tasks, assessment score, and an assessment score X study task interaction term. Results were presented both overall and by study task. Post hoc tests were conducted within the model to test whether the relationships differed across study tasks.

An a priori power analysis was completed using the key dependent variables in this study (H_f range and LD minima) using a published study with the same structure[8]. Assuming the sample size of 16 participants in the straight line gait and pre-planned turns tasks and 8 participants in the late-cued turns task yielded a power of at least 80% to detect a 1.5 standard deviation difference in these variables between experimental tasks (i.e., our primary analysis) while assuming a within-participant correlation of at least 0.5 and accommodating the Bonferroni corrected alpha necessary to maintain a 2-tailed familywise alpha of 0.05 ($p < 0.0167$) across the planned comparisons. For within-participant analyses, a minimum number of 10 trials per participant per study condition yielded 88.7% power to detect a 1.5 standard deviation difference in key study metrics between study tasks for the participant-level analysis, while accommodating the Bonferroni adjusted per-comparison alpha ($p < 0.0167$) necessary to maintain an overall alpha of 0.05 across the planned comparisons. A Bonferroni adjustment was chosen for the purposes of power analysis because it is highly conservative and the Holm test, which was used in all analyses, is based on the empirical p-value attained at the time of actual data analysis. (**Supplemental Document 1** includes full statistical result tables).

Results

Group-level results for H_f and LD across tasks are displayed in Fig. 3; Table 1 and participant-level results for H_f and LD across tasks are displayed in Fig. 4 and **Supplemental Document 1**, **Tables S2 and S3**.

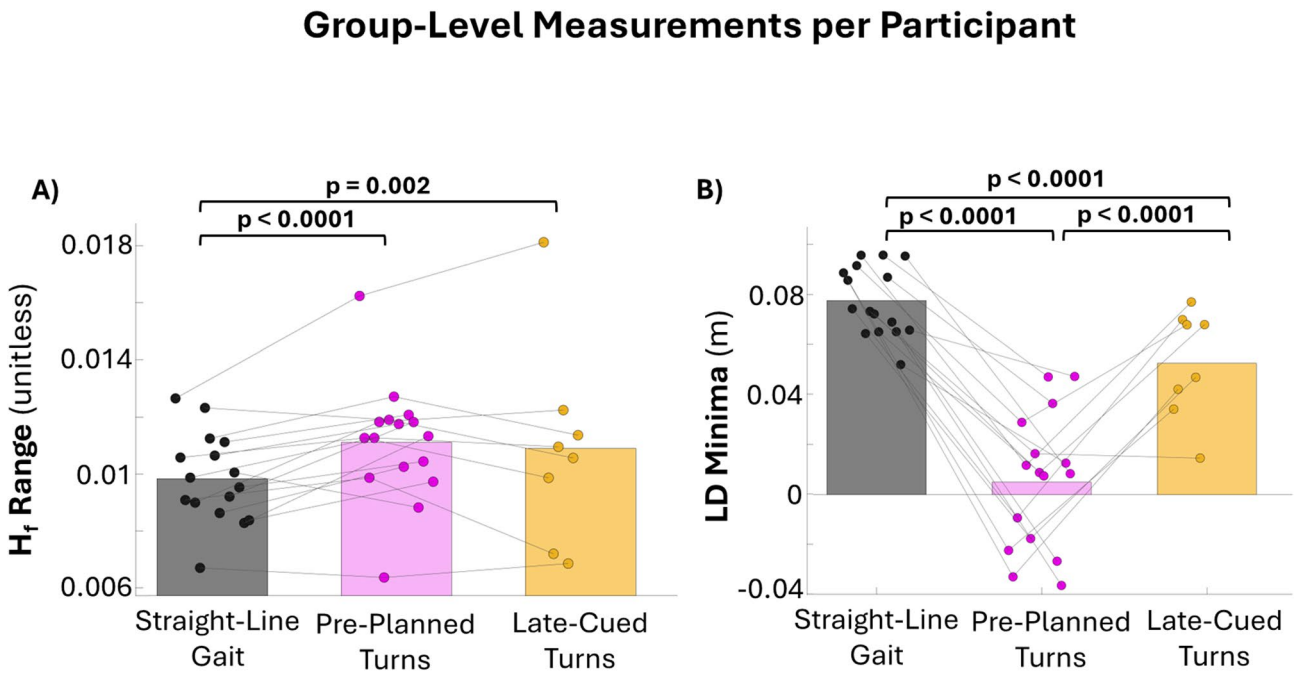

Fig. 3. Group-level statistical results for **A** angular momentum range (H_f) and **B** lateral distance minima (LD). Each participant's average H_f range and LD minima values across trials per task are indicated by dots and connected with lines. Group-level average per task indicated by bars. Brackets and p-values indicate significant differences.

Metric	Task			*p*-values		
	SLG Mean (95% CI)	PP Mean (95% CI)	LC Mean (95% CI)	SLG vs. PP	SLG vs. LC	PP vs. LC
H_fRange (unitless)	0.0098 (0.0094, 0.0102)	0.0111 (0.0107, 0.0114)	0.0109 (0.0103, 0.0114)	**< 0.0001**	**0.002**	0.59
LD Min (m)	0.0783 (0.0738, 0.0829)	0.0045 (-0.0001, 0.0090)	0.0540 (0.0474, 0.0607)	**< 0.0001**	**< 0.0001**	**< 0.0001**

Table 1. Group-level significant findings for H_f range and LD minima. Group level mean, confidence interval, and p-value for frontal plane angular momentum range, lateral distance minima across tasks: Straight-line gait (SLG), pre-planned turns (PP), and late-cued turns (LC). Bold p-values indicate significance.

Frontal plane angular momentum range

Using group-level statistical analyses, H_f range was smaller during straight-line gait than pre-planned ($p < 0.0001$) or late-cued turns ($p = 0.002$) (Table 1; Fig. 3). There was no significant difference between H_f range during pre-planned turns and late-cued turns at a group-level.

Participant-specific analyses revealed that not all participants followed the group-level statistical findings for H_f range comparisons across tasks (Fig. 4, Supplemental Document 1, Table S2). For the group finding that H_f range was smaller during straight-line gait than pre-planned turns, ten of 16 demonstrated this result statistically and one statistically demonstrated the opposite result. For the group finding that H_f range was smaller during straight-line gait than it was during late-cued turns, two participants demonstrated this result and two statistically demonstrated the opposite result. For the group finding that there was no statistical difference in H_f range between pre-planned and late-cued turns, six of eight participants demonstrated this result and two participants had larger H_f ranges during pre-planned turns than late-cued turns.

Lateral distance minima

Using group-level statistical analyses, LD minima was larger during straight-line gait than during either pre-planned ($p < 0.0001$) or late-cued ($p < 0.0001$) turns, and LD minima was larger during late-cued turns than during pre-planned turns ($p < 0.0001$) (Table 1; Fig. 3).

Participant-specific analyses revealed that all participants demonstrated the group finding of larger LD minima during straight-line gait than during pre-planned turns (Fig. 4, **Supplemental Document 1**, **Table S3**). Four of eight participants demonstrated the group finding of larger LD minima during straight-line gait than

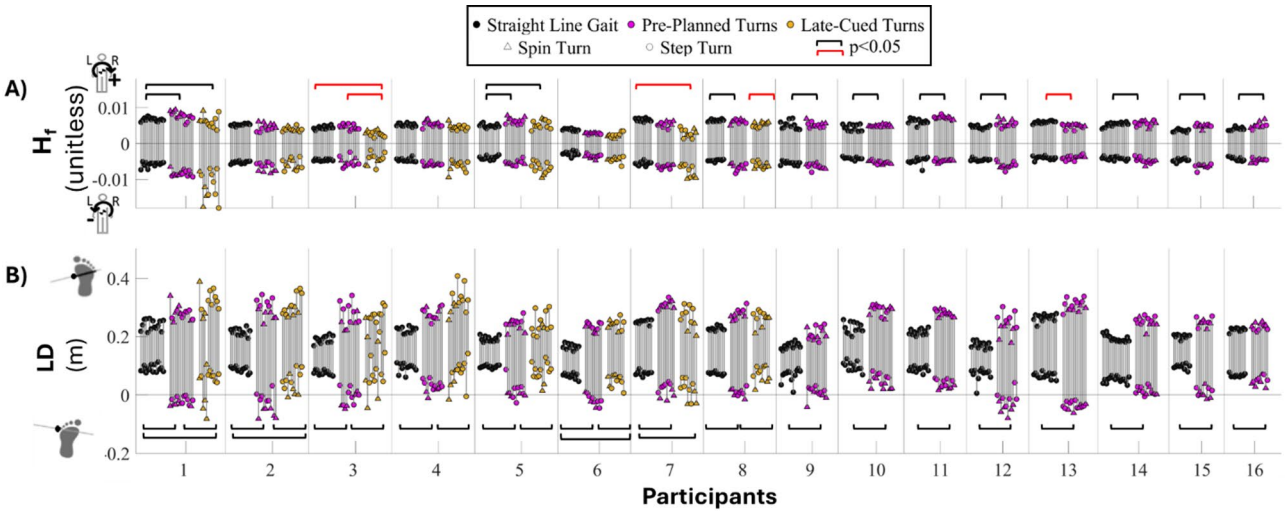

Fig. 4. Participant-level statistical results. Data points represent min, max and range for **A** H_f (positive values indicate clockwise rotation, when viewed posteriorly) and **B** LD for all trials across tasks and participants. Statistical analysis for H_f indicates changes in range, statistical significance for LD indicates changes in minima. Black bars indicate significant differences that align with the group-level significant findings, red bars indicate significant differences that oppose the group-level significant findings. Spin turns are indicated by triangles (when the left foot was closest to center of intersection) and step turns are indicated by circles (when the right foot was closest to center of intersection).

Metric	Task			*p*-values		
	SLG Mean (95% CI)	**PP** Mean (95% CI)	**LC** Mean (95% CI)	**SLG vs. PP**	**SLG vs. LC**	**PP vs. LC**
Gait Speed (m/s)	1.198 (1.182, 1.214)	1.003 (0.987, 1.020)	0.693 (0.669, 0.717)	**< 0.0001**	**< 0.0001**	**< 0.0001**
Stride Width (m)	0.0926 (0.0842, 0.1010)	0.1571 (0.1487, 0.1655)	0.1902 (0.1775, 0.2028)	**< 0.0001**	**< 0.0001**	**< 0.0001**
Stride Length (m)	1.2923 (1.2733, 1.3112)	1.1391 (1.1201, 1.1581)	0.8336 (0.8051, 0.8621)	**< 0.0001**	**< 0.0001**	**< 0.0001**
Step Length (m)	0.6456 (0.6364, 0.6548)	0.5771 (0.5679, 0.5863)	0.4345 (0.4211, 0.4478)	**< 0.0001**	**< 0.0001**	**< 0.0001**
Stride Duration (sec)	1.0724 (1.0578, 1.0870)	1.1245 (1.1099, 1.1392)	1.2235 (1.2016, 1.2455)	**< 0.0001**	**< 0.0001**	**< 0.0001**
Phase Duration (sec)	4.0984 (4.0259, 4.1708)	2.0997 (2.0271, 2.1724)	1.4902 (1.3849, 1.5955)	**< 0.0001**	**< 0.0001**	**< 0.0001**
Number of Footfalls	8.6368 (8.5146, 8.7591)	4.7513 (4.6287, 4.8739)	3.4444 (3.2668, 3.6221)	**< 0.0001**	**< 0.0001**	**< 0.0001**

Table 2. Group level statistics for additional contextual kinematic measures. Group level mean, confidence interval, and p-values for additional kinematic measures per each task phase of interest: straight line gait (SLG), pre-planned turns (PP), and late cued turns (LC). Bold p-values indicate significance.

during late-cued turns. Seven of eight participants demonstrated the group finding of larger LD minima during late-cued turns than during pre-planned turns.

Additional contextual kinematic measures

Using group-level statistical analysis, mean gait speed per task phase of interest was greater during straight-line gait than both pre-planned ($p < 0.0001$) and late-cued turns ($p < 0.0001$) and greater during pre-planned turns than late-cued turns ($p < 0.0001$) (Table 2). The effect of task on H_f range ($p = 0.02$) significantly differed by gait speed. The effect of task on LD minima did not significantly differ by gait speed ($p = 0.36$). Additional kinematic results are included in Table 2 and significantly differed across tasks ($p < 0.0001$). **Supplemental Document 1**, **Tables S4-S6** include additional gait speed and turn strategy context.

Exploration of possible associations with baseline assessments

Table 3 shows associations between baseline assessment scores and H_f range and LD minima. Generalized linear model results revealed that better scores (higher DGI, MOCA; lower FES-I, TUG, dual-TUG) for all baseline

Assessment	Angular Momentum Range (H_f)				Lateral Distance Minima (LD)				*p*-values: Slope between tasks		
	Combined Mean (p-value)	SLG Mean (CI) (p-value)	PP Mean (CI) (p-value)	LC Mean (CI) (p-value)	Combined Mean (p-value)	SLG Mean (CI) (p-value)	PP Mean (CI) (p-value)	LC Mean (CI) (p-value)	SLG vs. PP (H_f) (LD)	LG vs. LC (H_f) (LD)	PP vs. LC (H_f) (LD)
DGI	0.0006 (**<0.0001**)	0.0002 (−0.0001, 0.00046) (0.12)	0.0005 (0.0002, 0.0007) **(0.0003)**	0.0015 (0.0012, 0.0019) (**<0.0001**)	−0.001 (0.36)	0.0013 (−0.0023, 0.0048) (0.48)	−0.0035 (−0.0070, 0.0001) (0.054)	−0.0400 (−0.0086, 0.0007) (0.09)	(0.14) (0.19)	(**<0.0001**) (0.19)	(**<0.0001**) (0.87)
MOCA	0.0001 **(0.01)**	0.0001 (−0.00001, 0.0003) (0.07)	0.000048 (−0.0001, 0.0002) (0.54)	0.0003 (0.00003, 0.0005) **(0.03)**	−0.001 (0.13)	0.0010 (−0.0009, 0.0029) (0.31)	−0.0036 (−0.0055, −0.0016) **(0.0003)**	−0.0016 (−0.0044, 0.0012) (0.25)	(0.81) **(0.003)**	(0.81) (0.26)	(0.41) (0.26)
FES-I	−0.0001 **(0.002)**	−0.000003 (−0.0001, 0.0001) (0.92)	−0.00005 (−0.00001, 0.00002) (0.19)	−0.0008 (−0.0009, −0.0006) (**<0.0001**)	0.001 **(0.048)**	0.0002 (−0.0007, 0.0012) (0.63)	0.0016 (0.0007, 0.0025) **(0.0009)**	0.0023 (−0.00004, 0.0045) (0.054)	(0.39) (0.13)	(**<0.0001**) (0.21)	(**<0.0001**) (0.59)
TUG	−0.00048 (**<0.0001**)	−0.0002 (−0.0005, −0.00003) (0.08)	−0.0003 (−0.0005, −0.0001) **(0.04)**	−0.0028 (−0.0033, −0.0022) (**<0.0001**)	0.0047 **(0.006)**	−0.0017 (−0.0050, 0.0016) (0.31)	0.0102 (0.0069, 0.0135) (**<0.0001**)	0.0133 (0.0059, 0.0208) **(0.0005)**	(0.83) (**<0.0001**)	(**<0.0001**) **(0.0007)**	(**<0.0001**) (0.45)
Dual-TUG	−0.0002 (**<0.0001**)	−0.0001 (−0.0002, 0.00004) (0.20)	−0.0001 (−0.0002, 0.00004) (0.18)	−0.0010 (−0.0012, −0.0007) (**<0.0001**)	0.002 **(0.03)**	−0.0005 (−0.0021, 0.0011) (0.53)	0.0033 (0.0017, 0.0049) (**<0.0001**)	0.0039 (0.0010, 0.0067) **(0.008)**	(0.97) **(0.003)**	(**<0.0001**) **(0.02)**	(**<0.0001**) (0.76)

Table 3. Associations between baseline assessment scores and balance metrics across tasks. Associations between baseline assessment scores and angular momentum range and lateral distance minima for combined tasks and across tasks. Mean, confidence interval, and p-values recorded. Significant p-values in bold. Post-hoc comparisons of slopes between tasks reported for angular momentum and lateral distance, significant p-values in bold text.

assessments were associated with greater H_f range for combined tasks and during late-cued turns. Better DGI and TUG scores had significant associations with greater H_f range during pre-planned turns. A post-hoc analysis showed that associations with H_f range between DGI, FES-I, TUG, and dual-TUG were stronger for late-cued turns than either during pre-planned turns or during straight line gait.

Better scores for FES-I, TUG, and dual-TUG had significant associations with smaller LD minima for combined tasks. Better scores for MOCA, FES-I, TUG, and dual-TUG had significant associations with smaller LD minima during pre-planned turns. Better scores for TUG and dual-TUG had significant associations with smaller LD during late-cued turns. A post-hoc analysis showed that associations with LD minima between MOCA, TUG, and dual-TUG were stronger during pre-planned turns than straight line gait, and associations between TUG and dual-TUG were stronger during late-cued turns than straight line gait.

Discussion

The purpose of this study was to compare frontal-plane balance measures during walking and turning performed by healthy older adults. As hypothesized, as a group, older adults used larger H_f ranges during pre-planned and late-cued turns than during straight-line gait. Based on prior research with young adults, we hypothesized that older adults would have a larger H_f range during late-cued turns when compared to pre-planned turns. However, this was not supported in our cohort of eight older adults who performed late-cued turns; there was no significant difference in H_f range between pre-planned and late-cued turns at the group-level. Additionally, as hypothesized, older adults used larger LD minima during straight-line gait as compared to pre-planned or late-cued turns, and larger LD minima during late-cued turns than pre-planned turns. For both H_f range and LD minima, within participant statistical analyses revealed marked differences across individuals in this cohort of healthy older adults. Finally, an additional post-hoc exploration between participant baseline assessments and their balance metrics was included to better understand across-participant differences to guide future research.

For H_f, most participants followed the group-level statistical result that H_f range was smaller during straight-line gait than during either pre-planned or late-cued turns. This group-level result follows what has been found in younger adults performing the same task[8]. However, some older adults used smaller H_f ranges during turns than straight-line gait, which was an unexpected result with respect to results found with younger adults performing similar tasks[8].

Prior research suggests that larger H_f is associated with less controlled balance[24,28]. However, this study prompts the question of whether using larger H_f ranges in certain contexts can indicate a *greater* ability to navigate more complex balance tasks like turns, while smaller H_f ranges can be indicative of a balance-protective strategy[23,52,53]. This is suggested by the results of the exploration into how assessment scores related to H_f. Older adults who scored higher on the DGI and MOCA and lower on the FES-I, TUG, and dual-TUG (scores associated with better balance) had higher H_f ranges than those who had worse scores on these assessments. This suggests that older adults who would otherwise be expected to have "better" balance based on their baseline assessment scores used greater H_f ranges than older adults who would be expected to have "worse" balance. When comparing across tasks, these associations were more pronounced during turning tasks, and even more so during late-cued turns, which was the most challenging task in our study. This highlights the importance of

studying turns of varied difficulty towards understanding real-life implications of differing balance strategies across older adults.

As expected, older adults used smaller LD minima during pre-planned or late-cued turns than they did during straight-line gait, as well as larger LD minima during late-cued turns than during pre-planned turns. These findings align with prior research with younger adults completing the same tasks[8]. It is possible that more challenging turns lead to more protective and varied stepping and balance strategies and, thus, larger LD minima. This is additionally emphasized by the possible relationships between LD minima and baseline assessment scores. In our preliminary explorations, there were associations between better assessment scores (greater DGI or MOCA, smaller FES-I, TUG, dual-TUG) and smaller LD minima, suggesting that greater clinical, cognitive, or psychosocial scores were associated with a greater ability to use smaller LD when walking and turning. Taken together, this study's exploratory results suggest that older adults who score better on baseline assessment tests may safely navigate or utilize a larger range of balance states, emulating those used by younger adults. Conversely, older adults who score worse may limit themselves to balance states that are biomechanically safer[53].

Additional kinematic measures provide context for the H_f range and LD minima findings, especially gait speed. Turns were performed with slower gait speeds than straight-line gait, with late-cued turns using the slowest gait speed. There is a known tradeoff between gait speed and turn curvature, with slower gait speeds facilitating sharper turns[54–57], which was also observed in our study. As observed in young adults, pre-planned turns resembled circular gait, previously concluded to cause the COM to be situated closer to the lateral edge of the BOS[8,58], while late-cued turns better resembled sharp 90 degree turns (Fig. 5). This also allowed for faster gait speeds during pre-planned turns since the change of direction was gradual rather than abrupt. Similar to what was concluded by Tillman et al. (2022) with young adults, faster gait speeds can help account for more negative LD minima during pre-planned turns, as the body performs controlled forward and lateral falls[8]. Statistically controlling for mean gait speed by including it as a covariate did not change the group level findings of H_f range and LD minima across tasks (Supplemental Document 1, Table S1). There is preliminary evidence to suggest that gait speed may moderate the effect of task on H_f range, but not LD minima. If gait speed categories or thresholds (high vs. low) were established for turns, additional analysis could be completed to further investigate the effects of gait speed on the descriptive balance measures of interest. In the meantime, it is best to observe how behaviors may change (e.g., stepping strategies, etc.) with different gait speeds *experimentally*. This is prompted by recognizing the variety of footfall placements observed across turn types in this study and knowing that increased gait speed has been previously linked to increased[59,60], decreased[61], or unchanged H_f[62,63]. Using a slower, preferred pace during late-cued turns could afford healthy older adults a greater ability to control H_f and LD than if it were performed more quickly. If it were performed more quickly, vastly different spatiotemporal gait, transverse-plane momenta generation, and balance recovery strategies could be used that have not yet been observed. Therefore, follow-up studies should *experimentally* control for gait speed to better understand the impact of gait speed on gait behaviors including balance control measures.

Out of 16 participants, only two had fallen in the past 12 months with varied contexts. Participant six fell while hiking, while traversing small rocks at a slight elevation, and participant 15 tripped over a bike bump on the street. Both participants who had fallen previously commented that they are more careful now ("*It put some fear in me. I'm much more careful now*", "*I learned you have to look down.*"), and both had smaller H_f ranges than those that had not fallen ($p < 0.0001$). However, since there were only two participants, this is not conclusive, but helps guide future specific research into the psychosocial repercussions of concern levels and balance regulation

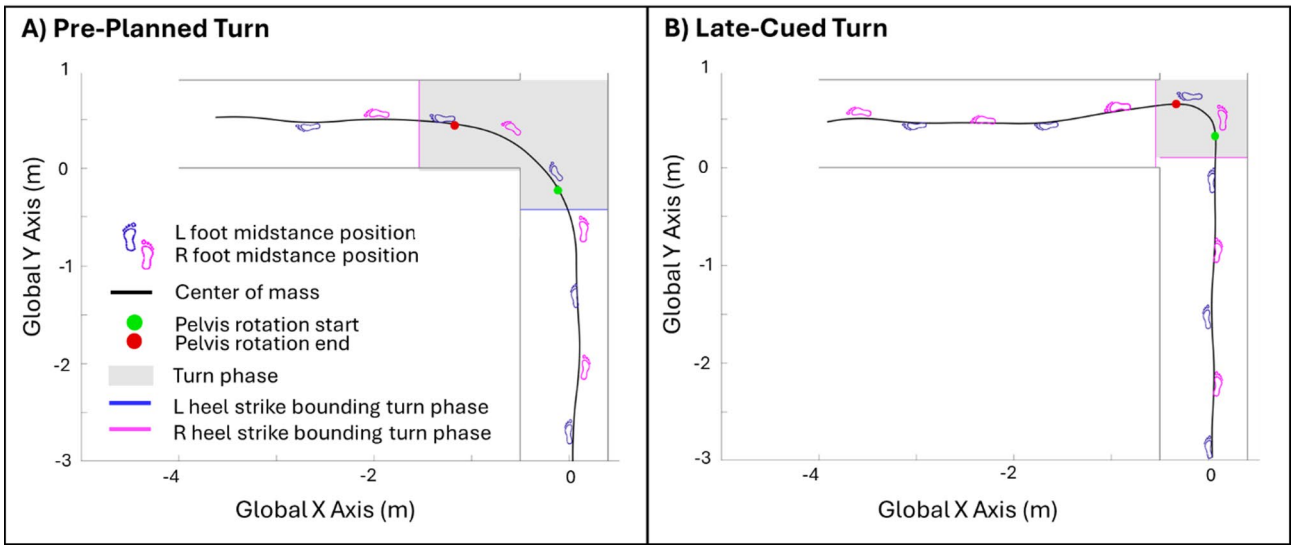

Fig. 5. Center of mass trajectory and footfall locations horizontal-plane plot examples of spin turns from participant one during **A** pre-planned and **B** late-cued turning tasks. Pre-planned turns often resemble circular gait with less apparent turn strategy characteristics.

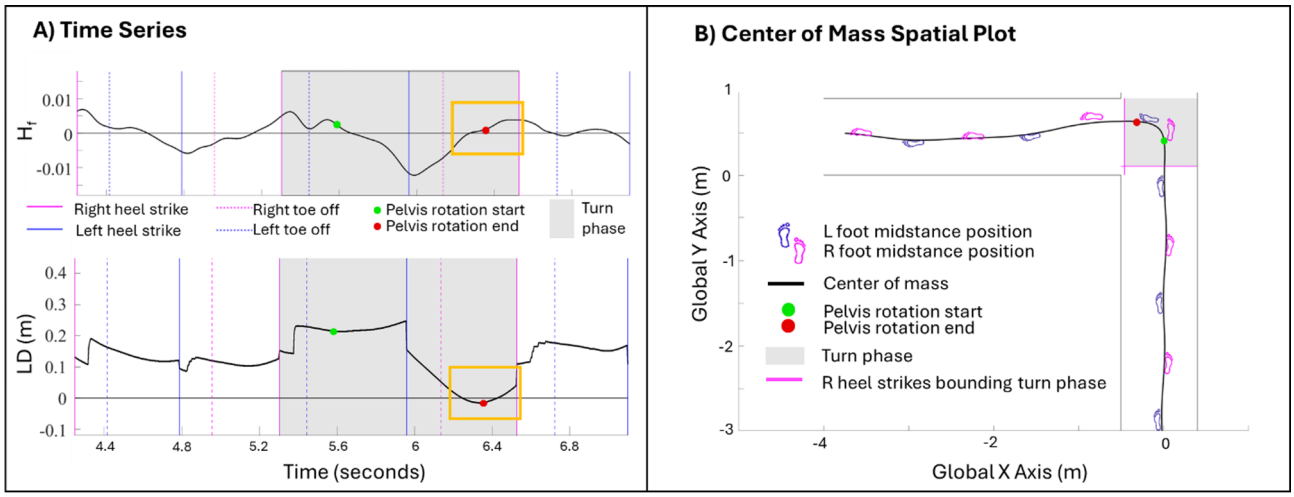

Fig. 6. Late-cued turn example from participant 1. **A** Time series of boxed trial for angular momentum (H_f) and lateral distance (LD). Negative LD and corresponding H_f indicated by the yellow squares. as LD reached negative values during the left turn, H_f returns to zero and begins to shift the body's H_f to the right. **B** Center of mass trajectory and footfall locations horizontal-plane plot of the example trial.

during walking and turning. The FES-I has demonstrated a robust ability to predict future falls[64], though these inquiries were not specific to biomechanics used during turning while walking.

While these frontal plane balance metrics provide important context into balance states experienced by older adults when walking and turning, dynamic balance is complex and includes concurrent interplay of multiple balance measures. As illustrated in the example timeseries in Fig. 6, H_f and LD extrema do not capture how each variable is controlled through time and in concert. For example, a large H_f maximum or minimum can be considered a less-precarious balance state if it occurs while LD is large and positive (i.e., the COM is situated within the BOS). Figure 6A shows the H_f and LD timeseries for an exemplar late-cued turning trial. By looking at the time point of the negative LD in the H_f timeseries plot, it is apparent that when LD is negative (i.e., the COM is left of the lateral edge of the left foot) H_f is increasing (i.e., a positive slope/differential indicating a positive moment applied about the COM), beginning to increase the body's momentum to the right, which coincides with returning the COM towards a large (more stable) LD value. However, at the time of the maximum negative H_f, the LD is positive, though it is about to decline towards its minima value. Building from the results from this study, future studies should seek a combined metric that takes the dynamic nature of both H_f and LD into account during each time point to better understand balance strategies older adults utilize when turning while walking. While Margin of Stability is often considered a metric to encompass rotational and positional balance control[10], pendular behavior required as an assumption has not been validated during turns, especially during more abrupt late-cued turns.

A major limitation in this paper (and related prior research) is in using extrema values of two balance metrics to quantify frontal plane dynamic balance control (as discussed in the prior paragraph). Additionally, the inclusion criteria yielded narrow (mostly healthy) ranges of baseline assessment scores; therefore, our explorations into how baseline assessments relate to turning biomechanics is only suggestive towards future research. This study uses a relatively small sample size, and the statistical analysis, though powerful, assumes that mean, variance, and covariance would not be appreciably different than if missing data (i.e., late-cued turns for eight participants) were present. Finally, this study contains only two male participants and two participants who had fallen previously, only left turns, and only eight participants who performed late-cued turns with the cues initiated manually. Thus, additional research can begin to explore effects of biological sex and fall history on balance control during turns, as well as more varied turns and automated cue timing.

Future research should continue to explore the relationships between biomechanical balance strategies used by older adults and their clinical, cognitive, and psychosocial assessments. Additionally, the dynamic way biomechanical metrics interact with one another through time during the turn should continue to be investigated.

Conclusions

On the group level, healthy older adults had larger H_f ranges during pre-planned and late-cued turns than during straight-line gait, but no difference in H_f ranges between pre-planned and late-cued turns. Additionally, older adults had smaller LD minima during pre-planned turns, followed by late-cued turns, and the largest LD minima during straight-line gait. These group trends indicate that different types of turning tasks may lead to different biomechanical strategies to safely complete a turn. Furthermore, within-participant statistical analyses revealed that several participants did not follow the group-level results, indicating that balance strategies varied across individuals. In an exploration of possible relationships between baseline assessments and frontal plane balance measures during walking and turning, results suggest that better clinical, cognitive, and psychosocial scores related to greater H_f ranges and smaller LD minima. These trends were more apparent during more difficult

turning tasks than they were during straight-line gait. These results motivate a more careful interpretation of H_f range and LD minima when it comes to inferring balance control *abilities*. In other words, trends that had been explained as indicative of poor balance control (e.g., larger H_f ranges) could be *viable* strategies used by both healthy young[8] and healthy older adults during turns.

Data availability

All data generated or analyzed during this study are included in this published article and its supplementary information files.

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Author contributions
ZH, MT, JML collected the data. ZH processed the data. ZH and AZ interpreted the data. JM performed statistical analysis. ZH wrote the manuscript, ZH and AZ edited it. AZ secured funding. All authors read and approved the final manuscript.

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Declarations

Ethics approval and consent to participate
All participants provided informed consent in accordance with the Stevens Institute of Technology

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Consent for publication
Not applicable.

Competing interests
The authors declare that they have no competing interests.

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